

EniacXI Hackathon • Team Socrates

Iris

Look. Link. Speak.

An eye-tracking wearable that turns where you look into what you want to say — giving voice to people with severe motor or speech impairment.

A 4-person build • ESP32-CAM + Cloud LLM



Iris

The problem

Half a million of people can think clearly but cannot move or speak to be understood.



— former VP of JD.com, diagnosed with ALS in 2019



Stephen Hawking — physicist, lived with ALS for 55 years



Your big, expressive eyes become the only window to the world.

The problem

Millions can think clearly but cannot move or speak to be understood.



Locked in, not switched off

Severe motor & speech impairment (ALS, late-stage cerebral palsy, brainstem stroke) **leaves cognition intact** but expression blocked.



Existing AAC is slow & costly

Dedicated gaze-speech devices can cost thousands, need calibrated screens, and tie the user to a fixed display.

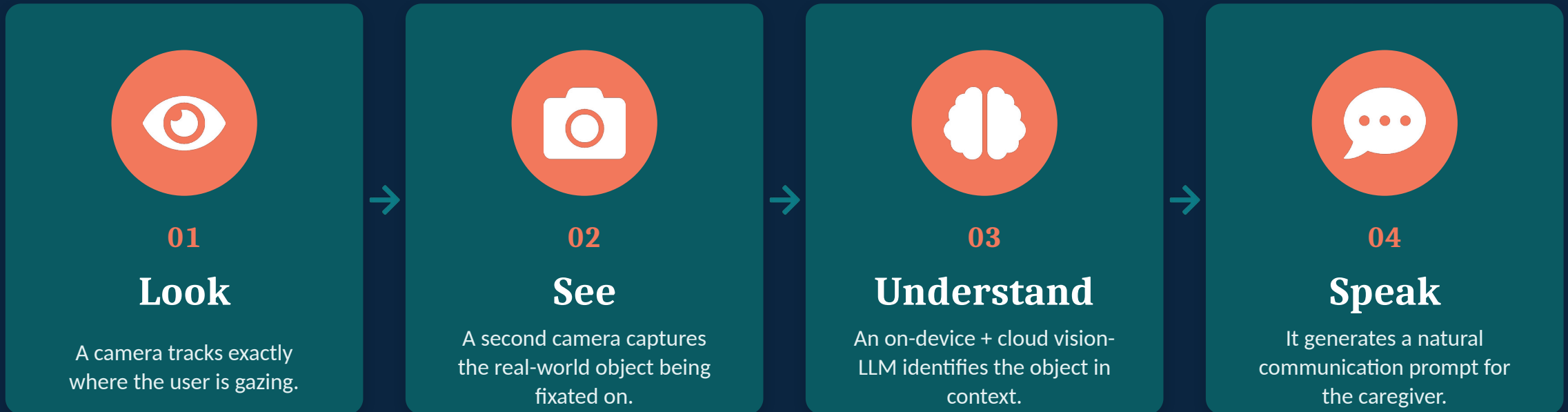


The world isn't on the screen

Real needs are physical — a cup, a door, a person. Screen-based boards can't point at what's actually in the room.

Our idea

Track the gaze. Read the scene. Let AI speak the intent.



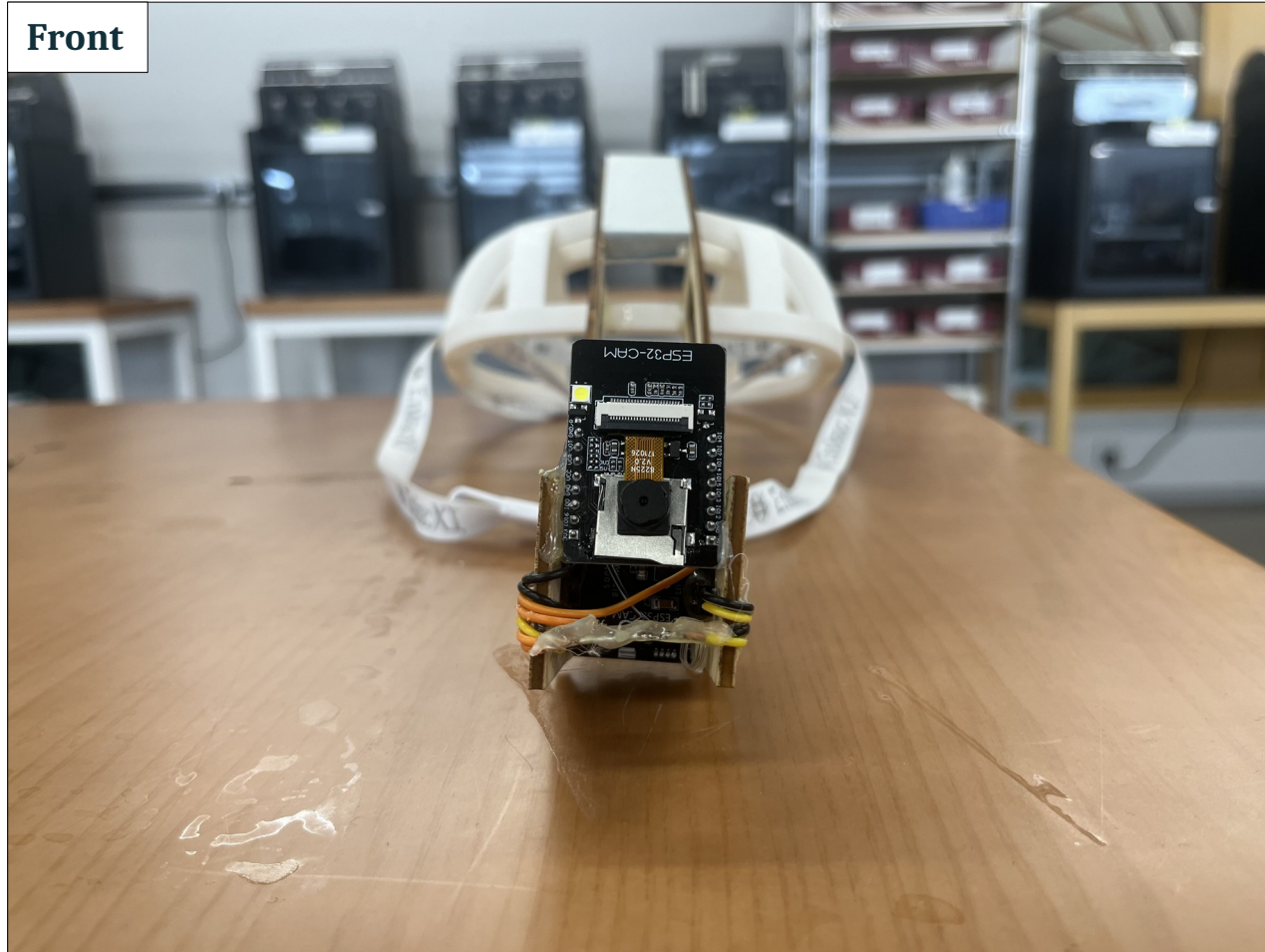
No screen to read. No buttons to press. The user just looks — the system does the rest.

Introducing,

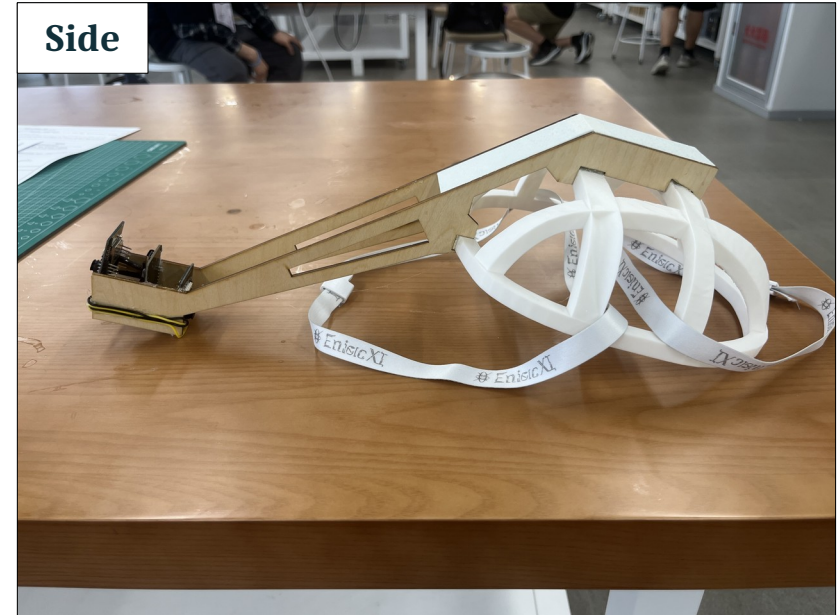
Iris



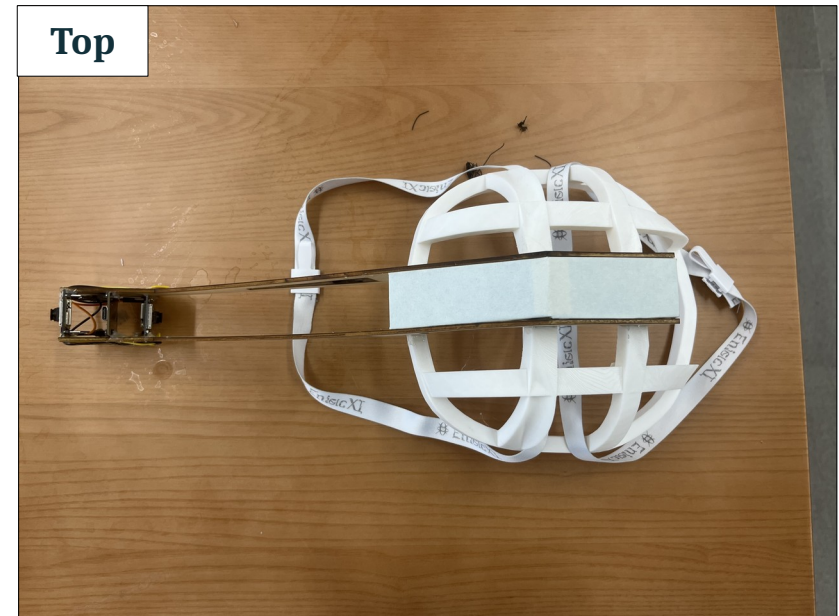
Front



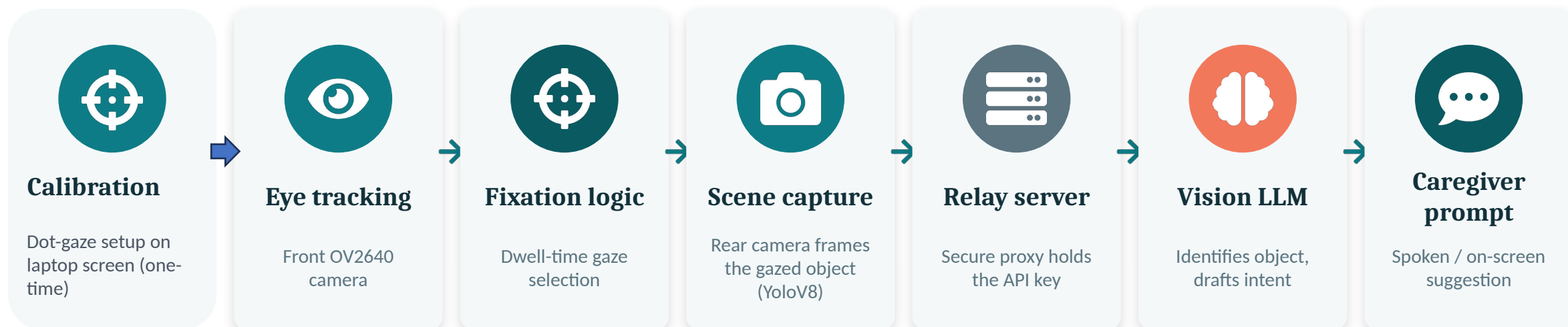
Side



Top



How it works — system pipeline



Key design choice: the LLM is triggered only on a stable gaze fixation (dwell-time gating) — not every frame. This controls latency, battery, and API cost.

How it works — Calibration



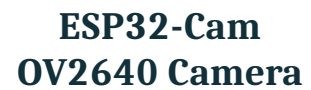
Calibration

Dot-gaze setup on
laptop screen (one-
time)

Point 1/9



Look at the dot, hold your head and the laptop steady



Front / IR camera +
ARKit face anchor

How it works — Fixation logic



Dwell time > eye clicks



Fixation logic

Dwell-time gaze
selection

How it works — Object Identification



YoloV8 identified a bottle from the rear camera view

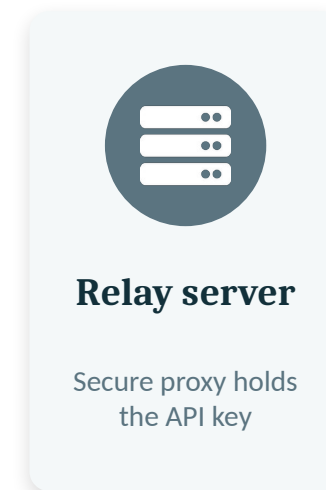
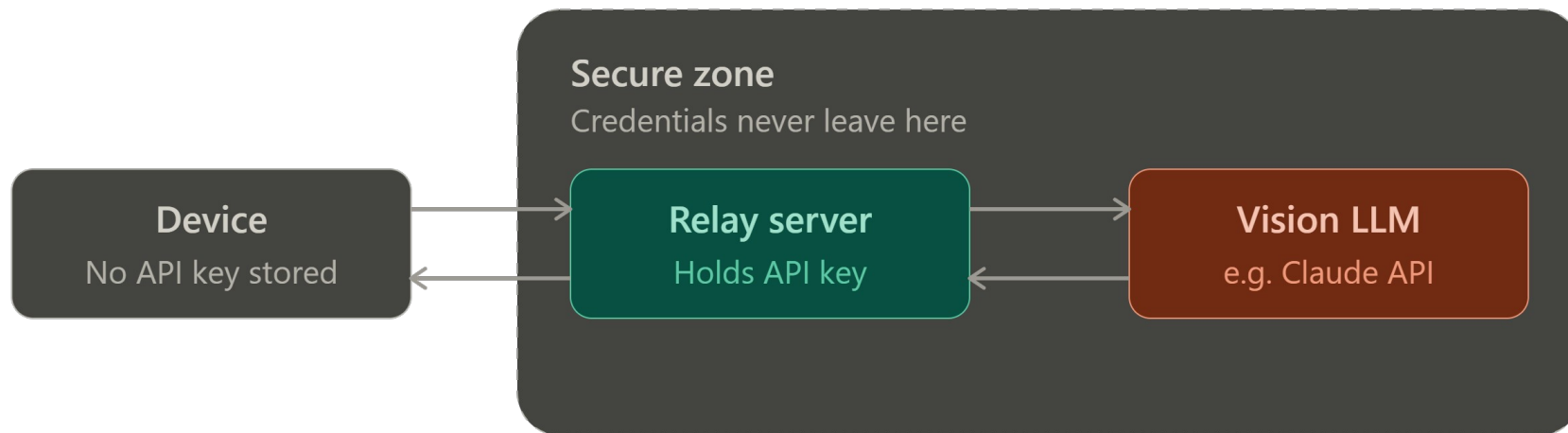


Eye tracking

Front / IR camera +
ARKit face anchor



How it works — Relay Server



How it works — Vision LLM



Haiku 4.5

fast, low-latency model (claude haiku) by design

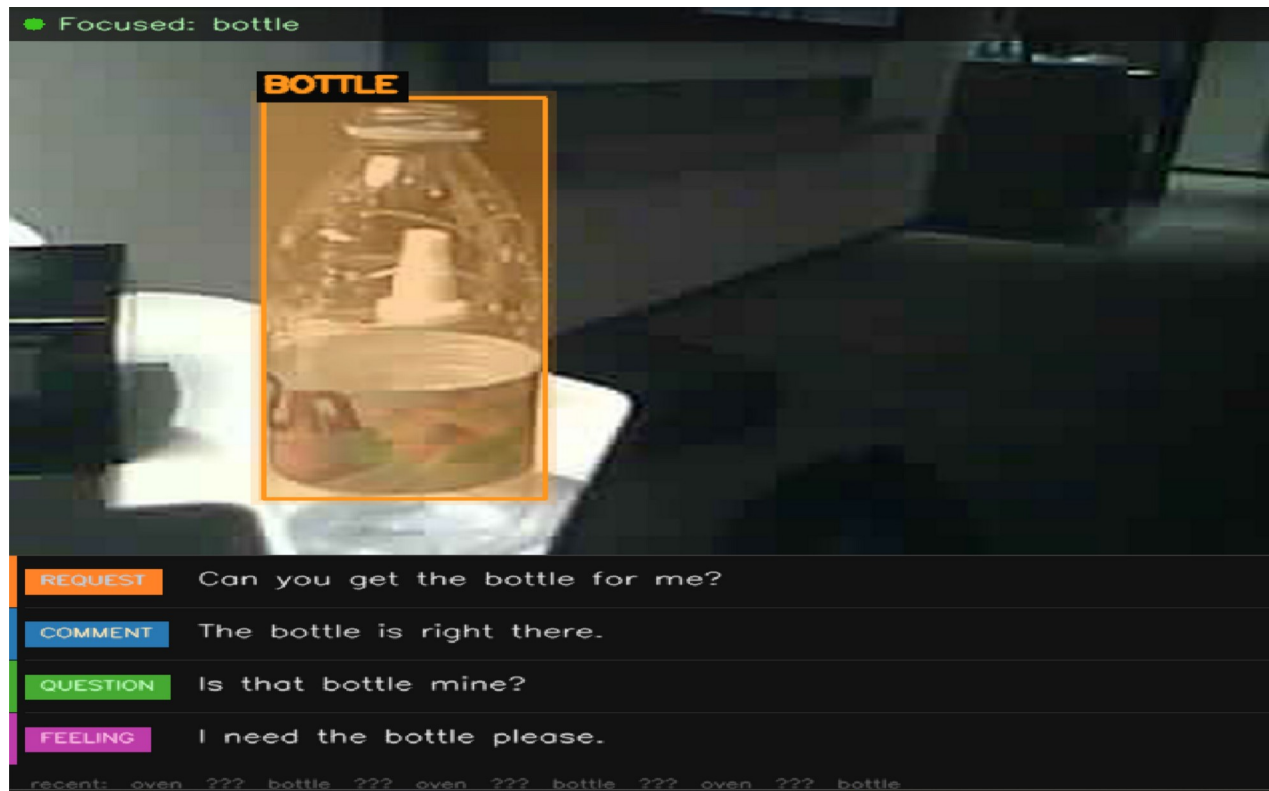


Vision LLM

Identifies object,
drafts intent

The offline template fallback is a great resilience story — if connectivity drops, the device degrades gracefully instead of going silent.

How it works — Caregiver Prompt



⚡ Reading out the most likely prompt based on context(Time, surrounding objects, preference, personal OBJECT, etc.



**Caregiver
prompt**

Spoken / on-screen
suggestion

The technical unlock

One Apple API runs world tracking and eye tracking at the same time, in one shared coordinate space.



ARWorldTrackingConfiguration

.userFaceTrackingEnabled



Rear-camera world tracking + front camera face/eye tracking, simultaneously



Both feeds live in one ARKit coordinate space — no manual dual-camera sync



Available iOS 13+ on A12 Bionic and newer chips



ARFaceAnchor exposes gaze direction we ray-cast into the scene

Why this matters

Custom dual-camera synchronization is the hard part of any gaze + scene system.

We solved it. That single flag removes weeks of low-level engineering and lets the team focus on the actual product: fixation logic, LLM prompting, and the user experience.

Two hardware paths

Validate the software on the simple path first, then upgrade the optics.



Plan 1 • MVP

iPhone-first

- ✓ Phone mounted on a boom in front of the user
- ✓ Front camera → eye tracking via native ARKit
- ✓ Rear camera → scene capture
- ✓ Zero custom electronics — pure software

Lower risk • proves the pipeline fast



Plan 2 • Custom optics

Wearable frame

- 3D-printed hollow eyewear frame
- Embedded ESP32-CAM eye-tracking modules
- IR illumination (IR-cut filter removed)
- True glasses form factor — wearable, hands-free

Higher fidelity • the product vision

Getting the geometry right

The Cameras comfortable viewing distance happen to agree.

300–350
mm

Eye-to-camera working distance — satisfies the TrueDepth sensor AND is comfortable to view

~7°

Rear-camera tilt angle to mimic human vision

1 fix

Extrinsic calibration in code corrects the offset — no hardware change needed



Implication: this points to a boom-arm / head-mount form factor rather than traditional glasses — which is exactly why we test an adjustable rig before committing to printed dimensions.

Testing Results

The Cameras comfortable viewing distance happen to agree.

85%

Gaze tracking accuracy

92%

Object detection accuracy

0.8s

Fixation dwell time

1.5s

End-to-end reaction
time

~\$40

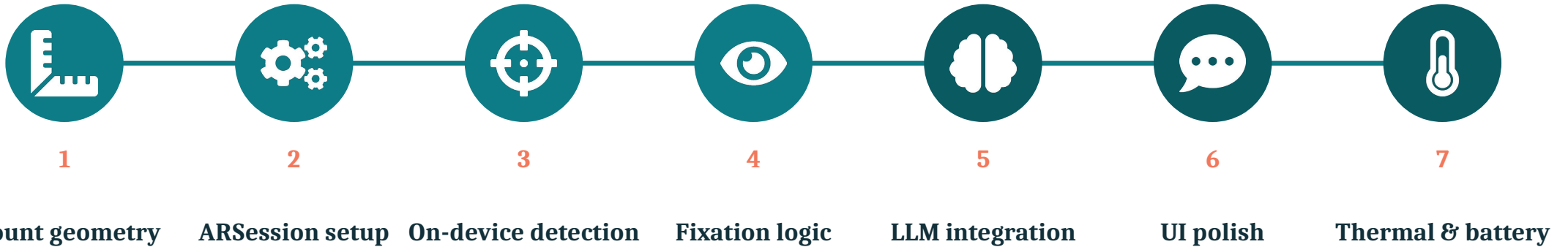
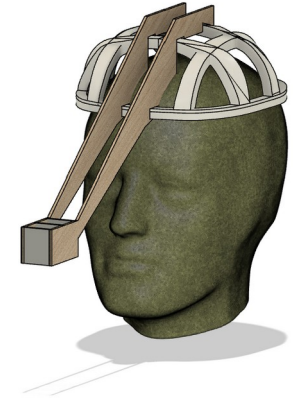
Total Cost



Sample size=20

Build order — de-risk in sequence

Validate the cheap, reversible things first; commit to fixed hardware last.



Adjustable rig: test distance & offset userFaceTrackingEnabled live Vision / CoreML object ID Dwell-time gaze selection Relay server → vision prompt Caregiver-facing prompts Sustained-use stress test

Principle: prove the software pipeline on the iPhone MVP before investing in custom optics — the lowest-risk path to a working demo.

Four people, four tracks

Parallel workstreams that converge at integration.



Eye camera & IR

IR-illuminated eye module,
filter removal, ESP32-CAM
tuning



Scene & networking

Rear-camera capture and the
relay-server data path



Fabrication

3D-printed frames and the
adjustable test rig



Software pipeline

ARKit, detection, fixation logic,
LLM, receiving UI

Software is the connective tissue — it's the track that turns four piles of hardware into one experience.

Risks & fallbacks

Every hard part has a planned escape hatch.



Custom frame too complex



Plan 2.5 — use off-the-shelf eye-tracking glasses instead of fully custom fabrication



Landscape tracking unstable



Switch to portrait + a clip-on wide-angle lens on the rear camera



LLM latency / cost too high



Dwell-time gating already throttles calls; tune the dwell threshold up



API key exposure



Relay server pattern — keys never live in the app binary

On the horizon

From hackathon prototype to a tool people can rely on.



Empirical rig testing

Measure mount distance & offset angle on an adjustable prototype before printing fixed dimensions.



Sequential integration

ARSession → detection → fixation → LLM → UI → thermal/battery, validated one stage at a time.



Plan 2.5 option

Off-the-shelf eye-tracking glasses as a faster route to a wearable form factor.



Beyond ALS

Test with target users to wear glasses to boost productivity



Iris

Team Socrates

Look. Link. Speak.

**Iris gives a voice back to people who still
have everything to say.**

Thank you — questions welcome.

EniacXI Hackathon • Keystone Academy